

Bharathwaj Nedoumaran

+1 260-410-8783 | bharathwaj.nedou@gmail.com | linkedin.com/in/bharathwaj-nedoumaran | github.com/Bharath-Maran
Problem Solving Critical Thinking Teamwork Leadership Quick Learner Keen Observer

EDUCATION

Purdue University <i>Master of Science in Computer Science</i>	Fort Wayne, Indiana, United States <i>August 2025 – May 2027</i>
St. Joseph's Institute of Technology <i>Bachelor of Engineering in Computer Science and Engineering</i>	Chennai, India <i>August 2019 – May 2023</i>

EXPERIENCE

Application Developer II <i>Oracle</i>	September 2023 – July 2025 <i>Bengaluru, India</i>
- Led Install, Security, and Performance testing across multi-platform environments (Windows, Linux, IBMi, Solaris, HP, and AIX) and databases (SQL, Oracle, DB2). - Automated JD Edwards workflows with Python Flask REST APIs, reducing environment maintenance and testing effort by 25%. - Built a Release Engineering Dashboard (Oracle APEX, JavaScript, SQL, Flask, Oracle REST Data Services) to integrate test data across 30+ projects, improving testing delivery by 40%.	
Software Engineer Intern <i>Aspire Systems</i>	March 2023 – August 2023 <i>Chennai, India</i>
- Trained in Python, Core/Advanced Java, SQL, Spring, Flask, OOP, Data Structures, and SDLC. - Developed two full-stack applications using Java (Spring), Python (Flask), SQL Server, HTML, CSS, and JavaScript showcasing end-to-end development skills. - Delivered a test-takers module for an LMS within 3 months (Agile), enhancing learning for 5,000+ employees and demonstrating strong end-to-end development capabilities.	

RESEARCH EXPERIENCE

Computer Science and Engineering, St. Joseph's Institute of Technology <i>Advisor: Prof. Mrs. K. Jaspin, and Prof. Mr. P. Mithun</i>	June 2021 – May 2023
- Gained 1.5 years of practical research experience in machine learning, beginning with mentorship from Prof. Mrs. K. Jaspin, resulting in a published paper. - Further refined my skills under Prof. P. Mithun, where I explored advanced machine learning techniques and contributed to significant research outcomes. - Learned to view challenges and setbacks as valuable lessons, strengthening my problem-solving abilities. - Improved teamwork dynamics, enabling efficient cooperation with peers to achieve high-quality results in our research projects.	

RESEARCH INTEREST

1. Machine Learning / Artificial Intelligence	2. Human Computer Interaction	3. Robotics
---	-------------------------------	-------------

PUBLICATIONS LIST

- B. Nedoumaran, A. M. Prosper, S. Maddu and M. P, "Precision Farming Using Machine Learning and Data Analytics," 2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSSES), Chennai, India, 2023.
- P. Aashik Mathew, A. Jagarlamudi, N. Bharathwaj and K. Jaspin, "Real-Time Number Plate and Helmet Detection of Motorcyclists using YOLOv5 and ResNet-50," 2023 International Conference on Sustainable Computing and Data Communication Systems (ICSCDS), Erode

PROJECTS

Precision Farming using Machine Learning and Data Analytics <i>St. Joseph's Institute of Technology</i>	December 2022 – March 2023
---	----------------------------

- Collaborated with a peer to collect and analyze extensive datasets of over 5,000+ samples and trained a machine learning model to predict crops using advanced algorithms.
- Achieved an outstanding 99.31% accuracy rate by rigorously testing and comparing six different supervised learning algorithms, utilizing XGBoost algorithm for optimal results, and improved efficiency.

Exploratory Analysis of Rainfall Data in India for Agriculture

July 2022 – November 2022

St. Joseph's Institute of Technology

- Led a team of 3 in developing a highly accurate rainfall prediction model for India using Linear Regression, achieving a remarkable 97% accuracy through advanced hyper-parameter tuning.
- Conducted in-depth analysis of a 10-year weather dataset, utilizing IBM Watson and other IBM Machine Learning tools to extract insights and enhance model robustness.
- Successfully deployed the model in a Flask environment, ensuring optimal scalability and performance under the mentorship of an IBM expert, demonstrating strong technical and project management skills.

Motorcyclists Helmet and License Plate Detection Using Yolo V5 and Resnet 50

January 2022 – June 2022

St. Joseph's Institute of Technology

- Co-developed a lightweight, real-time detection system (14MB model size) for motorcyclist helmet and license plate recognition, achieving 98.89% mAP, 94.6 F1-score, and processing speed of 130 FPS, ensuring high accuracy and efficiency.
- Utilized Canny Edge algorithm for frame preservation, enabling clear face and license plate identification, contributing to a comprehensive and scalable solution for real-time road safety monitoring.

TECHNICAL SKILLS

Programming Languages	Java, Python, SQL, PL/SQL, HTML, CSS.
Frameworks	Spring, Flask, Django.
Developer Tools	Git, Visual Studio Code, Eclipse, PyCharm.
Database Tools	MySQL, Oracle Database, MongoDB, and DB2.
Libraries	pandas, NumPy, Matplotlib.
Operating System	Windows, Linux, Solaris, AIX, IBMi, and HP.

ACADEMIC REFERENCES

Prof. Dr. J. Dafni Rose, Professor and Head
Department of Computer Science and Engineering
St. Joseph's Institute of Technology, OMR, Chennai - 600119
Phone: +91 99401 11231

Email: hodcsc@stjosephstechnology.ac.in

Prof. Mrs. K. Jaspin, Assistant Professor
Department of Computer Science and Engineering
St. Joseph's Institute of Technology, OMR, Chennai - 600119
Phone: +91 97910 87450

Email: jaspink@stjosephstechnology.ac.in

Prof. Mr. P. Mithun, Assistant Professor
Department of Computer Science and Engineering
St. Joseph's Institute of Technology, OMR, Chennai - 600119
Phone: +91 80721 67713

Email: mithunp@stjosephstechnology.ac.in

PROFESSIONAL REFERENCES

Mr. Maran Govindarajan, Senior Manager
Oracle, Bengaluru, India - 560035
Phone: +91 98459 81125

Email: maran.govindarajan@oracle.com

Mr. Subbu Sambandam, Associate Director – Talent Nurturing
Aspire Systems, Chennai, India - 603103
Phone: +91 98404 68855

Email: subbu.sambandam@aspiresys.com